

PROJECT REPORT

Automobile Price Driver Analysis & Market Segmentation (EDA)

201 Vehicles | 29 Features | 22 Makes | 1985 Ward's Automotive Yearbook
Python · pandas · seaborn · SQL | Portfolio EDA Project

Prepared by: Neela Vinay

neelavinni9@gmail.com | [linkedin.com/in/vinay-neela](https://www.linkedin.com/in/vinay-neela) | github.com/NeelaVinay

1. Executive Summary

What determines the price of a car? This EDA answers that question with data — not intuition. Analysing the 1985 Ward's Automotive Yearbook dataset across 201 vehicles, 29 features, and 22 makes, this project identifies which technical specifications most strongly predict vehicle price, how those drivers interact, and how the market naturally segments into four price tiers.

0.889 Engine Size → Price (r) Strongest single predictor	115% RWD Premium over FWD \$19,911 vs \$9,245 avg	31% Turbo Price Premium \$16,255 vs \$12,384 avg	2.9× High vs Low HP Ratio \$25,899 vs \$8,928 avg
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Three features — engine size ($r=0.889$), curb weight ($r=0.828$), and horsepower ($r=0.802$) — together account for over 80% of price variance. Drive wheels, body style, and aspiration type add discrete segment premiums on top. This has direct applications in pricing model development, insurance risk scoring, and inventory valuation.

2. Business Problem

Vehicle pricing is complex: two cars with the same sticker price may have radically different specifications, risk profiles, and market positions. This EDA was designed to answer five specific questions:

- Which single feature is the most reliable predictor of vehicle price?
- What premium do performance indicators (RWD, turbo, high HP) command?
- How do makes cluster into market segments — mass market vs luxury vs niche?
- What is the quantified relationship between fuel efficiency and price?
- How do body style and drive-wheel combinations create distinct value tiers?

3. Dataset Overview

Attribute	Value
Source	1985 Ward's Automotive Yearbook (UCI ML Repository)
Total Records	201 vehicles (196 after null removal)
Features	29 (physical specs, engine specs, insurance rating, fuel economy)
Unique Makes	22 (Toyota, Nissan, Honda, BMW, Mercedes-Benz, Jaguar, Porsche...)
Price Range	\$5,118 — \$45,400
Mean Price	\$13,207
Median Price	\$10,096

Attribute	Value
Null Handling	5 rows dropped (stroke: 4, horsepower-binned: 1) → 196 clean records
Feature Engineering	Horsepower binned: Low (<100hp) / Medium (100–150hp) / High (>150hp)

4. Methodology & Tool Stack

Tool	Role	Key Outputs
Python (pandas, NumPy)	Data cleaning, aggregation, correlation	Pearson r matrix, groupby summaries, outlier detection
Seaborn / Matplotlib	Statistical visualisation	Heatmap, regression plots, boxplots, distribution charts
SQL	Segmentation & validation	15 queries: price bands, segment pivot, feature ranking, outlier detection
Jupyter Notebook	End-to-end analytical narrative	Reproducible, documented EDA with narrative commentary

5. Key Findings

5.1 Feature Correlation with Price — Complete Ranking

Rank	Feature	Pearson r	r ²	Direction	Strength
1	engine-size	0.889	79.0%	+	Very Strong
2	curb-weight	0.828	68.6%	+	Very Strong
3	horsepower	0.802	64.3%	+	Very Strong
4	width	0.744	55.4%	+	Strong
5	highway-mpg	-0.707	50.0%	–	Strong (inverse)
6	city-mpg	-0.693	48.0%	–	Strong (inverse)
7	length	0.689	47.5%	+	Strong
8	wheel-base	0.592	35.0%	+	Moderate
9	bore	0.544	29.6%	+	Moderate
10	peak-rpm	-0.104	1.1%	–	Weak — not a reliable predictor
11	compression	0.068	0.5%	+	Negligible

Key insight: Peak RPM — which many buyers associate with performance — explains less than 2% of price variance. Compression ratio is essentially uncorrelated with price. These counterintuitive findings are analytically valuable precisely because they contradict buyer intuition.

5.2 Drive Wheels — 115% RWD Price Premium

Drive Type	Vehicle Count	Avg Price	Min Price	Max Price	vs FWD
RWD	70	\$19,911	\$8,495	\$45,400	+115% ↑
4WD	8	\$10,241	\$7,603	\$13,415	+11%
FWD	118	\$9,245	\$5,118	\$36,880	Baseline

RWD commands a \$10,666 structural premium over FWD regardless of brand. For pricing models, drive-wheels should be treated as a categorical premium flag, not a continuous predictor.

5.3 Body Style Price Analysis

Body Style	Count	Avg Price	Min	Max	Price Range Width
Hardtop	8	\$22,208	\$16,503	\$32,250	\$15,747
Convertible	6	\$21,890	\$11,259	\$35,056	\$23,797
Sedan	93	\$14,228	\$5,118	\$45,400	\$40,282 (widest)
Wagon	25	\$12,372	\$6,229	\$18,920	\$12,691
Hatchback	64	\$9,766	\$5,151	\$36,880	\$31,729

5.4 Turbo Aspiration — 31% Price Premium

Aspiration	Count	Avg Price	Avg HP	Avg Engine cc	Price Premium
Turbocharged	36	\$16,255	143 hp	150cc	+31% vs standard
Standard	160	\$12,384	97 hp	126cc	Baseline

5.5 Horsepower Tier Analysis

HP Tier	Count	Avg Price	Avg HP	Price Multiplier
High (>150hp)	23	\$25,899	190 hp	2.9× vs Low ↑
Medium (100–150hp)	61	\$15,917	116 hp	1.8×

HP Tier	Count	Avg Price	Avg HP	Price Multiplier
Low (<100hp)	112	\$8,928	75 hp	1.0× (baseline)

5.6 Engine Location — Rear-Engine Exotic Premium

Engine Location	Count	Avg Price	Avg HP	Notes
Rear	3	\$34,528	172 hp	All Porsche — treat as exotic flag
Front	193	\$12,761	99 hp	Standard fleet

5.7 Make Market Positioning

Make	Catalog Size	Avg Price	Market Tier
Jaguar	2	\$33,900	Luxury Niche
Mercedes-Benz	8	\$33,647	Luxury Niche
Porsche	4	\$31,400	Luxury Niche
BMW	10	\$26,119	Premium Volume
Volvo	11	\$18,063	Near-Premium
Toyota	32	\$13,495	Mass Market ← largest catalog
Nissan	18	\$13,284	Mass Market
Honda	13	\$8,872	Economy
Chevrolet	3	\$6,575	Economy

5.8 MPG vs Price Trade-off (r = -0.693)

City MPG Band	Vehicles	Avg Price	Avg MPG	Pattern
Very Low (<18)	~30	\$26,500	15 mpg	Luxury / performance
Low (18–24)	~60	\$15,200	21 mpg	Mid-range
Mid (25–34)	~80	\$9,800	29 mpg	Economy
High (35+)	~26	\$8,100	38 mpg	Micro / compact

Every 10 mpg increase in city economy correlates with approximately \$5,000 lower price. High-mpg vehicles are economy cars; low-mpg vehicles are luxury and performance. This is a direct, quantifiable consumer trade-off.

6. Four-Tier Market Segmentation

Tier	Criteria	Vehicles	Avg Price	Typical Brands
Tier 1 — Luxury / Performance	RWD + HP≥150 + Engine≥160cc	~23	\$30K+	BMW, Porsche, Jaguar, Mercedes
Tier 2 — Near-Luxury	RWD + (HP≥100 or Engine≥130cc)	~47	\$18–\$28K	Volvo, Audi, BMW entry
Tier 3 — Sporty Economy	FWD + HP≥100	~40	\$11–\$17K	Toyota, Honda performance trims
Tier 4 — Entry / Economy	All others	~91	\$6–\$11K	Honda, Chevrolet, Mitsubishi

7. Metric Proof Matrix

Every metric traceable to a specific query or formula. Reproducible from the public GitHub repository.

Metric	Value	SQL	Jupyter Notebook
Engine-size correlation	r=0.889	✓	✓
Curb-weight correlation	r=0.828	✓	✓
Horsepower correlation	r=0.802	✓	✓
City-mpg correlation	r=-0.693	✓	✓
RWD avg price	\$19,911	✓	✓
FWD avg price	\$9,245	✓	✓
RWD vs FWD premium	115%	✓	calc
Turbo avg price	\$16,255	✓	✓
Standard avg price	\$12,384	✓	✓
Turbo premium	31%	✓	calc
High HP avg price	\$25,899	✓	✓
Low HP avg price	\$8,928	✓	✓

Metric	Value	SQL	Jupyter Notebook
HP price ratio	2.9×	✓	calc
Price range	\$5,118–\$45,400	✓	✓
Rear-engine avg price	\$34,528	✓	✓
Null rows removed	5	n/a	✓

8. Business Applications & Recommendations

- Pricing model feature selection — use engine-size, curb-weight, horsepower as primary numeric predictors; add drive-wheels and aspiration as categorical flags. Drop peak-rpm ($r=-0.104$) and compression ratio ($r=0.068$) — they add noise, not signal.
- Insurance risk scoring — symboling and normalized-losses are more reliable insurance inputs than mechanical specs. Compression ratio is essentially uncorrelated with price, disqualifying it as an insurance proxy.
- Inventory valuation — hardtop and convertible body styles have the highest floor prices. Use body-style as a tier multiplier in used-car valuation models. Hatchbacks are the safe volume play but lowest margin per unit.
- Market entry strategy — only 3 rear-engine vehicles exist in this dataset, all Porsche, all >\$30K. Any new entrant in that segment competes against Porsche directly. FWD hatchback (Honda, Chevrolet) is high volume, low price, requiring scale economics.
- Consumer positioning — the \$5K-per-10-mpg relationship is a clear, quantifiable trade-off that marketing can use to justify premium pricing for low-mpg performance vehicles without competing purely on spec sheets.

GitHub: github.com/NeelaVinay/Automobile-Exploratory-Data-Analysis-EDA

This report was prepared as a portfolio deliverable demonstrating EDA capability across a structured multivariate dataset. All findings are reproducible from the public GitHub repository. SQL validation scripts are included alongside the Python notebook to demonstrate cross-tool analytical rigour.